

TABLE III (Continued)

## Iridium

| eV   | cm <sup>-1</sup> | μm     | <i>n</i> | <i>k</i> |
|------|------------------|--------|----------|----------|
| 2.1  | 16,940           | 0.5904 | 2.40     | 4.48     |
| 2.0  | 16,130           | 0.6199 | 2.50     | 4.57     |
| 1.9  | 15,320           | 0.6526 | 2.57     | 4.68     |
| 1.8  | 14,520           | 0.6888 | 2.64     | 4.81     |
| 1.7  | 13,710           | 0.7293 | 2.69     | 4.92     |
| 1.6  | 12,900           | 0.7749 | 2.68     | 5.08     |
| 1.5  | 12,100           | 0.8266 | 2.65     | 5.39     |
| 1.4  | 11,290           | 0.8856 | 2.72     | 5.74     |
| 1.3  | 10,490           | 0.9537 | 2.85     | 6.07     |
| 1.2  | 9,679            | 1.033  | 2.96     | 6.41     |
| 1.1  | 8,872            | 1.127  | 3.04     | 6.84     |
| 1.0  | 8,065            | 1.240  | 3.15     | 7.31     |
| 0.9  | 7,259            | 1.378  | 3.19     | 7.88     |
| 0.8  | 6,452            | 1.550  | 3.14     | 8.61     |
| 0.7  | 5,646            | 1.771  | 2.93     | 9.78     |
| 0.6  | 4,839            | 2.066  | 2.79     | 11.6     |
| 0.5  | 4,033            | 2.480  | 2.98     | 14.1     |
| 0.45 | 3,629            | 2.755  | 3.05     | 15.8     |
| 0.4  | 3,226            | 3.100  | 3.42     | 18.1     |
| 0.35 | 2,823            | 3.542  | 4.11     | 20.8     |
| 0.3  | 2,420            | 4.133  | 5.16     | 24.3     |
| 0.25 | 2,016            | 4.959  | 6.85     | 28.8     |
| 0.2  | 1,613            | 6.199  | 9.69     | 35.3     |
| 0.15 | 1,210            | 8.266  | 15.3     | 45.2     |
| 0.1  | 806.5            | 12.40  | 28.5     | 60.6     |

## MOLYBDENUM (Mo) VI

The optical properties of Mo [1–30] have been measured many times and the results generally agree rather well. Table IV lists the *n* and *k* values of Weaver *et al.* [19] and Henke *et al.* [31]. These data are plotted in Fig. 4 as smooth curves.

Weaver *et al.* [19] used samples spark-cut from Mo buttons of high purity that had been melted with an electron-beam gun. The samples were macro-etched, mechanically polished, electropolished, and annealed. After annealing, they were recleaned, electropolished, washed in acetone and then in ethyl alcohol, and transferred in air to the measuring apparatus. In the infrared region measurements were made by using a calorimetric technique [32], and at shorter wavelengths reflectance measurements at near-normal incidence

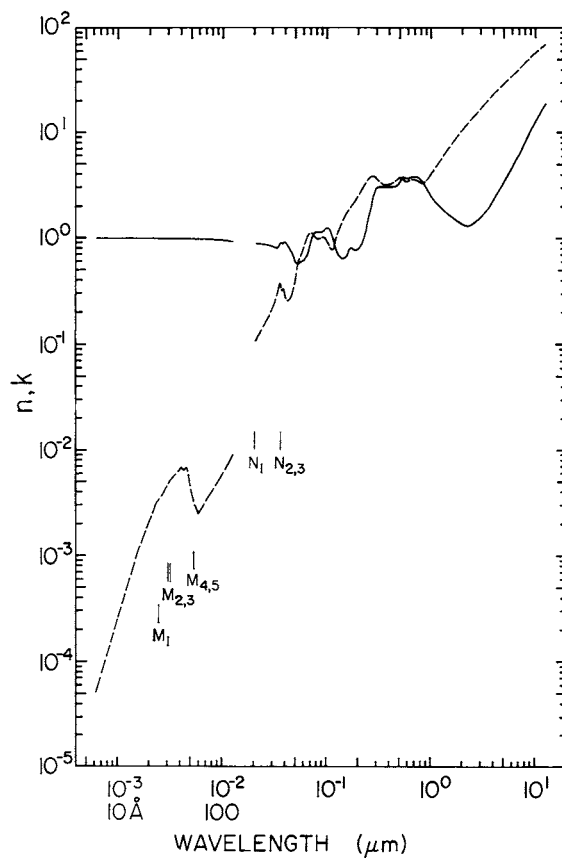
were made. These data were then analyzed by using the Kramers–Kronig method to obtain  $n$  and  $k$ .

There may be some influence of surface oxides on the vacuum ultraviolet data, but truly clean surfaces have not been measured, nor have oxidation studies been carried out. For wavelengths longer than about  $3\ \mu\text{m}$ , molybdenum resembles a free-electron metal, but, in principle, there are interband transitions at longer wavelengths that would affect the fit to a free-electron model for wavelengths shorter than about  $10\ \mu\text{m}$ .

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**Fig. 4.** Log-log plot of  $n$  (—) and  $k$  (---) versus wavelength in micrometers for molybdenum.

TABLE IV  
Values of  $n$  and  $k$  Obtained from Various References for Molybdenum<sup>a</sup>

| eV   | cm <sup>-1</sup> | Å     | $n$           | $k$                        |
|------|------------------|-------|---------------|----------------------------|
| 2000 |                  | 6.199 | 0.999622 [31] | 4.84x10 <sup>-5</sup> [31] |
| 1952 |                  | 6.351 | 0.999600      | 5.26                       |
| 1905 |                  | 6.508 | 0.999577      | 5.73                       |
| 1860 |                  | 6.665 | 0.999555      | 6.22                       |
| 1851 |                  | 6.831 | 0.999530      | 6.77                       |
| 1772 |                  | 6.997 | 0.999505      | 7.36                       |
| 1730 |                  | 7.166 | 0.999479      | 7.99                       |
| 1688 |                  | 7.345 | 0.999451      | 8.69                       |
| 1648 |                  | 7.523 | 0.999423      | 9.44x10 <sup>-5</sup>      |
| 1609 |                  | 7.705 | 0.999393      | 1.02x10 <sup>-4</sup>      |
| 1570 |                  | 7.897 | 0.999361      | 1.11                       |
| 1533 |                  | 8.087 | 0.999329      | 1.21                       |
| 1496 |                  | 8.287 | 0.999294      | 1.31                       |
| 1460 |                  | 8.492 | 0.999258      | 1.43                       |
| 1426 |                  | 8.694 | 0.999222      | 1.54                       |
| 1392 |                  | 8.906 | 0.999183      | 1.68                       |
| 1358 |                  | 9.129 | 0.999142      | 1.82                       |
| 1326 |                  | 9.350 | 0.999100      | 1.97                       |
| 1294 |                  | 9.581 | 0.999055      | 2.14                       |
| 1263 |                  | 9.816 | 0.999008      | 2.32                       |
| 1233 |                  | 10.05 | 0.99896       | 2.52                       |
| 1204 |                  | 10.30 | 0.99891       | 2.72                       |
| 1175 |                  | 10.55 | 0.99886       | 2.95                       |
| 1147 |                  | 10.81 | 0.99880       | 3.19                       |
| 1119 |                  | 11.08 | 0.99875       | 3.47                       |
| 1093 |                  | 11.34 | 0.99869       | 3.75                       |
| 1067 |                  | 11.62 | 0.99863       | 4.06                       |
| 1041 |                  | 11.91 | 0.99856       | 4.40                       |
| 1016 |                  | 12.20 | 0.99850       | 4.77                       |
| 992  |                  | 12.50 | 0.99843       | 5.15                       |
| 968  |                  | 12.81 | 0.99836       | 5.57                       |
| 945  |                  | 13.12 | 0.99829       | 6.02                       |
| 923  |                  | 13.43 | 0.99821       | 6.49                       |
| 901  |                  | 13.76 | 0.99813       | 7.01                       |
| 879  |                  | 14.10 | 0.99805       | 7.59                       |
| 858  |                  | 14.45 | 0.99797       | 8.19                       |
| 838  |                  | 14.79 | 0.99789       | 8.83                       |
| 818  |                  | 15.16 | 0.99781       | 9.53x10 <sup>-4</sup>      |
| 798  |                  | 15.54 | 0.99772       | 1.03x10 <sup>-3</sup>      |
| 779  |                  | 15.92 | 0.99764       | 1.11                       |
| 760  |                  | 16.31 | 0.99755       | 1.19                       |
| 742  |                  | 16.71 | 0.99747       | 1.27                       |
| 725  |                  | 17.10 | 0.99738       | 1.35                       |
| 707  |                  | 17.54 | 0.99728       | 1.44                       |
| 690  |                  | 17.97 | 0.99718       | 1.54                       |
| 674  |                  | 18.39 | 0.99708       | 1.64                       |
| 658  |                  | 18.84 | 0.99699       | 1.75                       |
| 642  |                  | 19.31 | 0.99688       | 1.87                       |
| 627  |                  | 19.77 | 0.99678       | 1.99                       |
| 612  |                  | 20.26 | 0.99668       | 2.13                       |
| 597  |                  | 20.77 | 0.99658       | 2.28                       |
| 583  |                  | 21.27 | 0.99649       | 2.43                       |

<sup>a</sup> The references from which the values were extracted are given in brackets.

TABLE IV (Continued)

| Molybdenum |                  |       |          |          |
|------------|------------------|-------|----------|----------|
| eV         | cm <sup>-1</sup> | Å     | <i>n</i> | <i>k</i> |
| 569        |                  | 21.79 | 0.99639  | 2.59     |
| 555        |                  | 22.34 | 0.99631  | 2.77     |
| 542        |                  | 22.87 | 0.99625  | 2.95     |
| 529        |                  | 23.44 | 0.99621  | 3.15     |
| 516        |                  | 24.03 | 0.99622  | 3.28     |
| 504        |                  | 24.60 | 0.99618  | 3.38     |
| 492        |                  | 25.20 | 0.99612  | 3.48     |
| 480        |                  | 25.83 | 0.99604  | 3.59     |
| 469        |                  | 26.43 | 0.99595  | 3.69     |
| 458        |                  | 27.07 | 0.99583  | 3.80     |
| 447        |                  | 27.74 | 0.99568  | 3.95     |
| 436        |                  | 28.44 | 0.99554  | 4.16     |
| 426        |                  | 29.10 | 0.99544  | 4.37     |
| 415        |                  | 29.87 | 0.99535  | 4.61     |
| 406        |                  | 30.54 | 0.99528  | 4.83     |
| 396        |                  | 31.31 | 0.99526  | 5.09     |
| 386        |                  | 32.12 | 0.99528  | 5.28     |
| 377        |                  | 32.89 | 0.99529  | 5.41     |
| 368        |                  | 33.69 | 0.99526  | 5.55     |
| 359        |                  | 34.53 | 0.99522  | 5.70     |
| 351        |                  | 35.32 | 0.99518  | 5.84     |
| 342        |                  | 36.25 | 0.99514  | 6.00     |
| 334        |                  | 37.12 | 0.99512  | 6.15     |
| 326        |                  | 38.03 | 0.99510  | 6.31     |
| 318        |                  | 38.99 | 0.99511  | 6.48     |
| 311        |                  | 39.86 | 0.99517  | 6.63     |
| 303        |                  | 40.92 | 0.99541  | 6.82     |
| 296        |                  | 41.88 | 0.99558  | 6.63     |
| 289        |                  | 42.90 | 0.99569  | 6.39     |
| 282        |                  | 43.96 | 0.99568  | 6.15     |
| 275        |                  | 45.08 | 0.99565  | 5.85     |
| 269        |                  | 46.09 | 0.99552  | 5.43     |
| 262        |                  | 47.32 | 0.99520  | 4.98     |
| 256        |                  | 48.43 | 0.99480  | 4.61     |
| 250        |                  | 49.59 | 0.99433  | 4.26     |
| 244        |                  | 50.81 | 0.99379  | 3.93     |
| 238        |                  | 52.09 | 0.99315  | 3.62     |
| 232        |                  | 53.44 | 0.99243  | 3.33     |
| 227        |                  | 54.62 | 0.99176  | 3.10     |
| 221        |                  | 56.10 | 0.99085  | 2.83     |
| 216        |                  | 57.40 | 0.9900   | 2.60     |
| 211        |                  | 58.76 | 0.9890   | 2.40     |
| 206        |                  | 60.18 | 0.9879   | 2.58     |
| 201        |                  | 61.68 | 0.9869   | 2.69     |
| 196        |                  | 63.25 | 0.9859   | 2.80     |
| 191        |                  | 64.91 | 0.9848   | 2.91     |
| 187        |                  | 66.30 | 0.9839   | 3.00     |
| 182        |                  | 68.12 | 0.9828   | 3.12     |
| 178        |                  | 69.65 | 0.9818   | 3.22     |
| 174        |                  | 71.25 | 0.9807   | 3.33     |
| 170        |                  | 72.93 | 0.9795   | 3.45     |
| 166        |                  | 74.69 | 0.9783   | 3.60     |

(continued)

TABLE IV (Continued)

## Molybdenum

| eV    | cm <sup>-1</sup> | Å       | <i>n</i>   | <i>k</i>              |
|-------|------------------|---------|------------|-----------------------|
| 162   |                  | 76.53   | 0.9770     | 3.75                  |
| 158   |                  | 78.47   | 0.9757     | 3.92                  |
| 154   |                  | 80.51   | 0.9742     | 4.10                  |
| 150   |                  | 82.65   | 0.9726     | 4.28                  |
| 147   |                  | 84.34   | 0.9713     | 4.42                  |
| 143   |                  | 86.70   | 0.9695     | 4.62                  |
| 140   |                  | 88.56   | 0.9680     | 4.78                  |
| 136   |                  | 91.16   | 0.9659     | 5.01                  |
| 133   |                  | 93.22   | 0.9641     | 5.19                  |
| 130   |                  | 95.37   | 0.9623     | 5.38                  |
| 127   |                  | 97.62   | 0.9603     | 5.59                  |
| 124   |                  | 99.98   | 0.9581     | 5.80                  |
| 121   |                  | 102.5   | 0.9557     | 6.03                  |
| 118   |                  | 105.1   | 0.9532     | 6.28                  |
| 115   |                  | 107.8   | 0.9504     | 6.54                  |
| 112   |                  | 110.7   | 0.9473     | 6.90                  |
| 110   |                  | 112.7   | 0.9451     | 7.19                  |
| 107   |                  | 115.9   | 0.9415     | 7.64                  |
| 104   |                  | 119.2   | 0.9377     | 8.14                  |
| 102   |                  | 121.5   | 0.9349     | 8.50                  |
| 100   |                  | 124.0   | 0.9318     | 8.89x10 <sup>-3</sup> |
| eV    | cm <sup>-1</sup> | μm      | <i>n</i>   | <i>k</i>              |
| 60.0  |                  | 0.02066 | 0.897 [19] | 0.106 [10]            |
| 50.0  |                  | 0.02480 | 0.868      | 0.151                 |
| 45.0  |                  | 0.02755 | 0.850      | 0.187                 |
| 40.0  |                  | 0.03100 | 0.828      | 0.235                 |
| 39.5  |                  | 0.03139 | 0.823      | 0.241                 |
| 39.0  |                  | 0.03179 | 0.818      | 0.250                 |
| 38.5  |                  | 0.03220 | 0.813      | 0.261                 |
| 38.0  |                  | 0.03263 | 0.810      | 0.273                 |
| 37.75 |                  | 0.03284 | 0.809      | 0.281                 |
| 37.5  |                  | 0.03306 | 0.808      | 0.289                 |
| 37.25 |                  | 0.03328 | 0.809      | 0.297                 |
| 37.0  |                  | 0.03351 | 0.810      | 0.303                 |
| 36.75 |                  | 0.03374 | 0.811      | 0.312                 |
| 36.5  |                  | 0.03397 | 0.812      | 0.321                 |
| 36.25 |                  | 0.03420 | 0.815      | 0.331                 |
| 36.0  |                  | 0.03444 | 0.819      | 0.341                 |
| 35.8  |                  | 0.03463 | 0.825      | 0.350                 |
| 35.6  |                  | 0.03483 | 0.831      | 0.358                 |
| 35.4  |                  | 0.03502 | 0.841      | 0.367                 |
| 35.2  |                  | 0.03522 | 0.853      | 0.373                 |
| 35.0  |                  | 0.03542 | 0.867      | 0.375                 |
| 34.8  |                  | 0.03563 | 0.880      | 0.373                 |
| 34.6  |                  | 0.03583 | 0.893      | 0.368                 |
| 34.4  |                  | 0.03604 | 0.901      | 0.360                 |
| 34.2  |                  | 0.03625 | 0.907      | 0.352                 |
| 34.0  |                  | 0.03647 | 0.911      | 0.344                 |
| 33.8  |                  | 0.03668 | 0.912      | 0.334                 |
| 33.6  |                  | 0.03690 | 0.910      | 0.326                 |
| 33.5  |                  | 0.03701 | 0.907      | 0.322                 |

TABLE IV (Continued)

| Molybdenum |                  |         |          |          |
|------------|------------------|---------|----------|----------|
| eV         | cm <sup>-1</sup> | μm      | <i>n</i> | <i>k</i> |
| 33.4       |                  | 0.03712 | 0.903    | 0.321    |
| 33.2       |                  | 0.03734 | 0.897    | 0.323    |
| 33.0       |                  | 0.03757 | 0.896    | 0.327    |
| 32.8       |                  | 0.03780 | 0.898    | 0.330    |
| 32.6       |                  | 0.03803 | 0.901    | 0.331    |
| 32.4       |                  | 0.03827 | 0.907    | 0.331    |
| 32.2       |                  | 0.03850 | 0.912    | 0.327    |
| 32.0       |                  | 0.03875 | 0.915    | 0.323    |
| 31.75      |                  | 0.03905 | 0.918    | 0.315    |
| 31.5       |                  | 0.03936 | 0.919    | 0.307    |
| 31.25      |                  | 0.03968 | 0.918    | 0.298    |
| 31.0       |                  | 0.04000 | 0.915    | 0.288    |
| 30.75      |                  | 0.04032 | 0.909    | 0.279    |
| 30.5       |                  | 0.04065 | 0.902    | 0.271    |
| 30.25      |                  | 0.04099 | 0.892    | 0.263    |
| 30.0       |                  | 0.04133 | 0.879    | 0.259    |
| 29.75      |                  | 0.04168 | 0.868    | 0.256    |
| 29.5       |                  | 0.04203 | 0.855    | 0.257    |
| 29.25      |                  | 0.04239 | 0.844    | 0.256    |
| 29.0       |                  | 0.04275 | 0.832    | 0.259    |
| 28.75      |                  | 0.04313 | 0.822    | 0.261    |
| 28.5       |                  | 0.04350 | 0.811    | 0.263    |
| 28.25      |                  | 0.04389 | 0.799    | 0.265    |
| 28.0       |                  | 0.04428 | 0.788    | 0.269    |
| 27.75      |                  | 0.04468 | 0.776    | 0.272    |
| 27.5       |                  | 0.04509 | 0.764    | 0.276    |
| 27.25      |                  | 0.04550 | 0.750    | 0.281    |
| 27.0       |                  | 0.04592 | 0.734    | 0.289    |
| 26.75      |                  | 0.04635 | 0.721    | 0.298    |
| 26.5       |                  | 0.04679 | 0.706    | 0.308    |
| 26.25      |                  | 0.04723 | 0.694    | 0.319    |
| 26.0       |                  | 0.04769 | 0.680    | 0.329    |
| 25.8       |                  | 0.04806 | 0.666    | 0.341    |
| 25.6       |                  | 0.04843 | 0.656    | 0.353    |
| 25.4       |                  | 0.04881 | 0.645    | 0.367    |
| 25.2       |                  | 0.04920 | 0.636    | 0.380    |
| 25.0       |                  | 0.04959 | 0.624    | 0.393    |
| 24.8       |                  | 0.04999 | 0.612    | 0.410    |
| 24.6       |                  | 0.05040 | 0.603    | 0.429    |
| 24.4       |                  | 0.05081 | 0.596    | 0.447    |
| 24.2       |                  | 0.05123 | 0.588    | 0.468    |
| 24.0       |                  | 0.05166 | 0.583    | 0.491    |
| 23.8       |                  | 0.05209 | 0.581    | 0.511    |
| 23.6       |                  | 0.05254 | 0.579    | 0.532    |
| 23.4       |                  | 0.05299 | 0.579    | 0.553    |
| 23.2       |                  | 0.05344 | 0.579    | 0.575    |
| 23.0       |                  | 0.05391 | 0.582    | 0.595    |
| 22.8       |                  | 0.05438 | 0.586    | 0.614    |
| 22.6       |                  | 0.05486 | 0.589    | 0.633    |
| 22.4       |                  | 0.05535 | 0.594    | 0.650    |
| 22.2       |                  | 0.05585 | 0.596    | 0.668    |
| 22.0       |                  | 0.05636 | 0.602    | 0.685    |

(continued)

TABLE IV (Continued)

## Molybdenum

| eV   | cm <sup>-1</sup> | μm      | <i>n</i> | <i>k</i> |
|------|------------------|---------|----------|----------|
| 21.8 |                  | 0.05687 | 0.607    | 0.700    |
| 21.6 |                  | 0.05740 | 0.608    | 0.715    |
| 21.4 |                  | 0.05794 | 0.609    | 0.733    |
| 21.2 |                  | 0.05848 | 0.610    | 0.752    |
| 21.0 |                  | 0.05904 | 0.612    | 0.773    |
| 20.8 |                  | 0.05961 | 0.616    | 0.794    |
| 20.6 |                  | 0.06019 | 0.621    | 0.815    |
| 20.4 |                  | 0.06078 | 0.626    | 0.838    |
| 20.2 |                  | 0.06138 | 0.632    | 0.862    |
| 20.0 |                  | 0.06199 | 0.639    | 0.887    |
| 19.8 |                  | 0.06262 | 0.648    | 0.913    |
| 19.6 |                  | 0.06326 | 0.660    | 0.940    |
| 19.4 |                  | 0.07391 | 0.674    | 0.968    |
| 19.2 |                  | 0.06458 | 0.691    | 0.996    |
| 19.0 |                  | 0.06526 | 0.712    | 1.02     |
| 18.8 |                  | 0.06595 | 0.737    | 1.05     |
| 18.6 |                  | 0.06666 | 0.766    | 1.08     |
| 18.4 |                  | 0.06738 | 0.799    | 1.10     |
| 18.2 |                  | 0.06812 | 0.833    | 1.11     |
| 18.0 |                  | 0.06888 | 0.866    | 1.12     |
| 17.8 |                  | 0.06965 | 0.903    | 1.13     |
| 17.6 |                  | 0.07045 | 0.940    | 1.14     |
| 17.4 |                  | 0.07126 | 0.978    | 1.14     |
| 17.2 |                  | 0.07208 | 1.01     | 1.13     |
| 17.0 |                  | 0.07293 | 1.04     | 1.12     |
| 16.8 |                  | 0.07380 | 1.07     | 1.11     |
| 16.6 |                  | 0.07469 | 1.10     | 1.10     |
| 16.4 |                  | 0.07560 | 1.12     | 1.08     |
| 16.2 |                  | 0.07653 | 1.13     | 1.06     |
| 16.0 |                  | 0.07749 | 1.14     | 1.04     |
| 15.8 |                  | 0.07847 | 1.15     | 1.02     |
| 15.6 |                  | 0.07948 | 1.15     | 1.01     |
| 15.4 |                  | 0.08051 | 1.15     | 0.994    |
| 15.2 |                  | 0.08157 | 1.14     | 0.989    |
| 15.0 |                  | 0.08266 | 1.14     | 0.987    |
| 14.8 |                  | 0.08377 | 1.14     | 0.987    |
| 14.6 |                  | 0.08492 | 1.13     | 0.990    |
| 14.4 |                  | 0.08610 | 1.13     | 0.997    |
| 14.2 |                  | 0.08731 | 1.14     | 1.01     |
| 14.0 |                  | 0.08856 | 1.15     | 1.01     |
| 13.8 |                  | 0.08984 | 1.15     | 1.02     |
| 13.6 |                  | 0.09117 | 1.17     | 1.02     |
| 13.4 |                  | 0.09253 | 1.18     | 1.02     |
| 13.2 |                  | 0.09393 | 1.20     | 1.03     |
| 13.0 |                  | 0.09537 | 1.21     | 1.02     |
| 12.8 |                  | 0.09686 | 1.23     | 1.01     |
| 12.6 |                  | 0.09840 | 1.24     | 0.993    |
| 12.4 |                  | 0.09999 | 1.25     | 0.975    |
| 12.2 |                  | 0.1016  | 1.26     | 0.953    |
| 12.0 |                  | 0.1033  | 1.26     | 0.921    |
| 11.8 |                  | 0.1051  | 1.25     | 0.886    |
| 11.6 |                  | 0.1069  | 1.23     | 0.845    |



TABLE IV (Continued)

## Molybdenum

| eV   | cm <sup>-1</sup> | μm     | <i>n</i> | <i>k</i> |
|------|------------------|--------|----------|----------|
| 11.4 |                  | 0.1088 | 1.18     | 0.802    |
| 11.2 |                  | 0.1107 | 1.12     | 0.782    |
| 11.0 |                  | 0.1127 | 1.05     | 0.774    |
| 10.8 |                  | 0.1148 | 0.977    | 0.791    |
| 10.6 |                  | 0.1170 | 0.909    | 0.835    |
| 10.4 |                  | 0.1192 | 0.859    | 0.881    |
| 10.2 |                  | 0.1216 | 0.812    | 0.930    |
| 10.0 |                  | 0.1240 | 0.770    | 0.987    |
| 9.8  |                  | 0.1265 | 0.735    | 1.05     |
| 9.6  |                  | 0.1291 | 0.707    | 1.12     |
| 9.4  |                  | 0.1319 | 0.685    | 1.19     |
| 9.2  |                  | 0.1348 | 0.670    | 1.25     |
| 9.0  |                  | 0.1378 | 0.652    | 1.33     |
| 8.8  |                  | 0.1409 | 0.647    | 1.41     |
| 8.6  |                  | 0.1442 | 0.649    | 1.49     |
| 8.4  |                  | 0.1476 | 0.659    | 1.57     |
| 8.2  |                  | 0.1512 | 0.672    | 1.65     |
| 8.0  |                  | 0.1550 | 0.685    | 1.73     |
| 7.9  |                  | 0.1569 | 0.697    | 1.77     |
| 7.8  |                  | 0.1590 | 0.711    | 1.81     |
| 7.7  |                  | 0.1610 | 0.726    | 1.85     |
| 7.6  |                  | 0.1631 | 0.745    | 1.90     |
| 7.5  |                  | 0.1653 | 0.781    | 1.93     |
| 7.4  |                  | 0.1675 | 0.808    | 1.95     |
| 7.3  |                  | 0.1698 | 0.816    | 1.96     |
| 7.2  |                  | 0.1722 | 0.813    | 1.98     |
| 7.1  |                  | 0.1746 | 0.808    | 2.00     |
| 7.0  |                  | 0.1771 | 0.798    | 2.04     |
| 6.9  |                  | 0.1797 | 0.791    | 2.08     |
| 6.8  |                  | 0.1823 | 0.784    | 2.12     |
| 6.7  |                  | 0.1851 | 0.777    | 2.18     |
| 6.6  |                  | 0.1879 | 0.779    | 2.24     |
| 6.5  |                  | 0.1907 | 0.783    | 2.30     |
| 6.4  |                  | 0.1937 | 0.789    | 2.36     |
| 6.3  |                  | 0.1968 | 0.798    | 2.43     |
| 6.2  |                  | 0.2000 | 0.813    | 2.50     |
| 6.1  | 49,200           | 0.2033 | 0.836    | 2.57     |
| 6.0  | 48,390           | 0.2066 | 0.850    | 2.64     |
| 5.9  | 47,590           | 0.2101 | 0.873    | 2.72     |
| 5.8  | 46,780           | 0.2138 | 0.895    | 2.80     |
| 5.7  | 45,970           | 0.2175 | 0.924    | 2.89     |
| 5.6  | 45,170           | 0.2214 | 0.959    | 2.99     |
| 5.5  | 44,360           | 0.2254 | 1.00     | 3.09     |
| 5.4  | 43,550           | 0.2296 | 1.06     | 3.20     |
| 5.3  | 42,750           | 0.2339 | 1.13     | 3.31     |
| 5.2  | 41,940           | 0.2384 | 1.22     | 3.42     |
| 5.1  | 41,130           | 0.2431 | 1.33     | 3.53     |
| 5.0  | 40,330           | 0.2480 | 1.46     | 3.62     |
| 4.9  | 39,520           | 0.2530 | 1.61     | 3.70     |
| 4.8  | 38,710           | 0.2583 | 1.75     | 3.76     |
| 4.7  | 37,910           | 0.2638 | 1.90     | 3.81     |
| 4.6  | 37,100           | 0.2695 | 2.06     | 3.84     |

(continued)

TABLE IV (Continued)

## Molybdenum

| eV   | cm <sup>-1</sup> | μm     | <i>n</i> | <i>k</i> |
|------|------------------|--------|----------|----------|
| 4.5  | 36,290           | 0.2755 | 2.21     | 3.87     |
| 4.4  | 35,490           | 0.2818 | 2.39     | 3.88     |
| 4.3  | 34,680           | 0.2883 | 2.59     | 3.86     |
| 4.2  | 33,880           | 0.2952 | 2.77     | 3.77     |
| 4.1  | 33,070           | 0.3024 | 2.91     | 3.67     |
| 4.0  | 32,260           | 0.3100 | 3.01     | 3.51     |
| 3.9  | 31,460           | 0.3179 | 3.04     | 3.40     |
| 3.8  | 30,650           | 0.3263 | 3.04     | 3.31     |
| 3.7  | 29,840           | 0.3351 | 3.04     | 3.27     |
| 3.6  | 29,040           | 0.3444 | 3.05     | 3.24     |
| 3.5  | 28,230           | 0.3542 | 3.06     | 3.21     |
| 3.4  | 27,420           | 0.3647 | 3.06     | 3.19     |
| 3.3  | 26,620           | 0.3757 | 3.06     | 3.18     |
| 3.2  | 25,810           | 0.3875 | 3.05     | 3.18     |
| 3.1  | 25,000           | 0.4000 | 3.03     | 3.22     |
| 3.0  | 24,200           | 0.4133 | 3.04     | 3.27     |
| 2.9  | 23,390           | 0.4275 | 3.05     | 3.33     |
| 2.8  | 22,580           | 0.4428 | 3.08     | 3.42     |
| 2.7  | 21,780           | 0.4592 | 3.13     | 3.51     |
| 2.6  | 20,970           | 0.4769 | 3.22     | 3.61     |
| 2.5  | 20,160           | 0.4959 | 3.36     | 3.73     |
| 2.4  | 19,360           | 0.5166 | 3.59     | 3.78     |
| 2.3  | 18,550           | 0.5391 | 3.79     | 3.61     |
| 2.2  | 17,740           | 0.5636 | 3.76     | 3.41     |
| 2.1  | 16,940           | 0.5904 | 3.68     | 3.45     |
| 2.0  | 16,130           | 0.6199 | 3.68     | 3.52     |
| 1.9  | 15,320           | 0.6526 | 3.74     | 3.58     |
| 1.8  | 14,520           | 0.6888 | 3.81     | 3.58     |
| 1.7  | 13,710           | 0.7293 | 3.84     | 3.51     |
| 1.6  | 12,900           | 0.7749 | 3.77     | 3.41     |
| 1.5  | 12,100           | 0.8266 | 3.53     | 3.30     |
| 1.4  | 11,290           | 0.8856 | 3.15     | 3.40     |
| 1.3  | 10,490           | 0.9537 | 2.77     | 3.74     |
| 1.2  | 9,679            | 1.033  | 2.44     | 4.22     |
| 1.1  | 8,872            | 1.127  | 2.16     | 4.85     |
| 1.0  | 8,065            | 1.240  | 1.94     | 5.58     |
| 0.9  | 7,259            | 1.378  | 1.74     | 6.48     |
| 0.86 | 6,936            | 1.442  | 1.70     | 6.90     |
| 0.82 | 6,614            | 1.512  | 1.64     | 7.35     |
| 0.78 | 6,291            | 1.590  | 1.60     | 7.83     |
| 0.74 | 5,968            | 1.675  | 1.52     | 8.38     |
| 0.70 | 5,646            | 1.771  | 1.48     | 8.99     |
| 0.66 | 5,323            | 1.879  | 1.43     | 9.67     |
| 0.62 | 5,001            | 2.000  | 1.38     | 10.4     |
| 0.58 | 4,678            | 2.138  | 1.34     | 11.3     |
| 0.54 | 4,355            | 2.296  | 1.35     | 12.4     |
| 0.50 | 4,033            | 2.480  | 1.37     | 13.5     |
| 0.46 | 3,710            | 2.695  | 1.46     | 14.9     |
| 0.42 | 3,388            | 2.952  | 1.57     | 16.5     |
| 0.38 | 3,065            | 3.263  | 1.70     | 18.4     |
| 0.34 | 2,742            | 3.647  | 1.99     | 20.8     |
| 0.30 | 2,420            | 4.133  | 2.44     | 23.8     |
| 0.29 | 2,339            | 4.275  | 2.58     | 24.7     |
| 0.28 | 2,258            | 4.428  | 2.74     | 25.6     |

TABLE IV (Continued)

| Molybdenum |                  |       |          |          |
|------------|------------------|-------|----------|----------|
| eV         | cm <sup>-1</sup> | μm    | <i>n</i> | <i>k</i> |
| 0.27       | 2,178            | 4.592 | 2.92     | 26.6     |
| 0.26       | 2,097            | 4.769 | 3.14     | 27.6     |
| 0.25       | 2,016            | 4.959 | 3.36     | 28.8     |
| 0.24       | 1,936            | 5.166 | 3.61     | 30.0     |
| 0.23       | 1,855            | 5.391 | 3.92     | 31.3     |
| 0.22       | 1,774            | 5.636 | 4.26     | 32.7     |
| 0.21       | 1,694            | 5.904 | 4.65     | 34.3     |
| 0.20       | 1,613            | 6.109 | 5.10     | 36.0     |
| 0.19       | 1,532            | 6.526 | 5.61     | 37.8     |
| 0.18       | 1,452            | 6.888 | 6.21     | 39.9     |
| 0.17       | 1,371            | 7.293 | 6.90     | 42.2     |
| 0.16       | 1,291            | 7.749 | 7.74     | 44.7     |
| 0.15       | 1,210            | 8.266 | 8.78     | 47.5     |
| 0.14       | 1,129            | 8.856 | 9.96     | 50.7     |
| 0.13       | 1,049            | 9.537 | 11.4     | 54.4     |
| 0.12       | 967.9            | 10.33 | 13.4     | 58.4     |
| 0.11       | 887.2            | 11.27 | 15.6     | 63.1     |
| 0.10       | 806.5            | 12.40 | 18.5     | 68.5     |

## NICKEL (Ni) VII

Overall agreement among the published *n* and *k* spectra of Ni [1–42] is good. Table V lists *n* and *k* values chosen and the pertinent references. These data are plotted in Fig. 5 as smooth curves.

The data extending from 1.3 to 124 Å are those of Zombeck *et al.* [40] determined as described in the introduction. Data calculated from a model of Henke *et al.* [43] overlap in the 6.1- to 124- Å region. The data from 88 to 248 Å are those of Bartlett *et al.* [42] (extinction coefficient) and Feldkamp *et al.* [39] (index of refraction). The data from 248 to 495 Å (Weaver *et al.* [41]), from 495 to 3540 Å (Vehse and Arakawa [13]), and from 4132 Å to 15.5 μm (Lynch *et al.* [17]) were combined for a Kramers–Kronig (KK) analysis, and the results are listed in the table and shown in Fig. 5 for 248 Å to 12.4 μm. The data of Henke [41] agree well with those of Feldkamp *et al.* [39] and Bartlett and Olson [42] in the overlap region.

Bartlett and Olson [42] used evaporated Ni films to obtain an absorption spectrum of Ni. They evaporated Ni onto thin-film substrates of carbon (1600 Å thick) or aluminum (900 Å thick). The pressure during deposition was less than 10<sup>-8</sup> torr. The Ni was of high purity (resistivity ratio of 100 to 200).